

Predictive Modeling Using Machine Learning

Introduction

Machine Learning enables computers to learn patterns from data and make predictions. This project predicts whether a student passes based on study hours.

Objective

To build and evaluate a machine learning model using student study-hour data.

Dataset Description

The dataset contains Hours Studied and Passed (0=Fail, 1=Pass).

Algorithm Used

Decision Tree Classifier was used to learn patterns and predict outcomes.

Methodology

Loaded data, split into training/testing sets, trained the model, tested predictions, and evaluated accuracy.

Tools Used

Python, Pandas, Scikit-learn, CSV Dataset.

Conclusion

The model successfully demonstrated predictive modeling and supervised learning concepts.